

Autonomous Vehicle Lane Following (EKF): Fusion of IMU and Camera/GPS Sensors

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Abstract—Autonomous vehicle lane following necessitates robust state estimation in the presence of nonlinear dynamics, asynchronous multi-rate sensors, and non-Gaussian measurement artifacts. This paper presents a comprehensive Extended Kalman Filter (EKF) architecture designed to fuse high-frequency Inertial Measurement Unit (IMU) kinematics (100 Hz) with lower-frequency Global Positioning System (GPS) (10 Hz) and Camera (30 Hz) observations. A 6-Degree-of-Freedom (DOF) state vector is employed to accurately estimate not only the planar position, velocity, and heading but also the critical slow-varying IMU hardware biases. To ensure filter stability against multipath errors, frame drops, and environmental outliers, a Normalized Innovation Squared (NIS) gating technique is integrated across all observation models. The estimated states are actively fed into a closed-loop Stanley Controller to achieve stable lateral lane keeping and longitudinal speed regulation without relying on ground truth data. Extensive Monte Carlo simulations (N=50) rigorously validate the statistical consistency and robustness of the proposed framework, demonstrating highly reliable dead reckoning capabilities during simulated GPS tunnel dropouts, robust bias estimation, and trajectory tracking bounded strictly within a 95% confidence interval.

Index Terms—Extended Kalman Filter, Sensor Fusion, Autonomous Navigation, Stanley Controller, NIS Gating, Kinematic Bicycle Model, Multi-Rate Estimation, Bias Tracking, Dead Reckoning.

I. INTRODUCTION

The rapid advancement of Advanced Driver Assistance Systems (ADAS) and fully autonomous vehicles is heavily reliant on the precision, continuity, and robustness of real-time state estimation algorithms. Autonomous lane following is a safety-critical application where relying on a single sensor modality is practically insufficient due to inherent physical limitations and volatile environmental constraints.

Cameras, utilizing computer vision, directly detect lane markings and provide critical cross-track error (CTE) and relative heading measurements. However, they are highly susceptible to sudden illumination changes, sun glare, shadows, false object detections, and varying weather conditions. The Global Positioning System (GPS) provides absolute global positioning, eliminating unbounded drift; nevertheless, it suffers from low update rates, non-Gaussian multipath errors in urban

environments, and complete signal loss (outages) in urban canyons or tunnels. Conversely, an Inertial Measurement Unit (IMU) provides high-frequency, high-bandwidth kinematic data (acceleration and yaw rate) that is immune to external environmental conditions. However, numerical integration of IMU data inevitably accumulates severe drift over time due to unmodeled, slow-varying hardware biases and thermal noise.

To overcome these distinct individual shortcomings, multi-sensor fusion utilizing recursive Bayesian estimation is the industry standard. By combining the absolute references of GPS and Camera systems with the high-frequency continuity of the IMU, a highly resilient navigation solution can be achieved.

This paper outlines the mathematical framework, detailed sensor characterization, and algorithmic implementation of an asynchronous, multi-rate Extended Kalman Filter (EKF) architecture. Coupled with a closed-loop Stanley Controller, the system is designed to track and control the vehicle dynamically, mimicking a true autonomous pipeline. The performance is rigorously evaluated using comprehensive single-run metrics and Monte Carlo statistical analysis, with a specific focus on filter consistency via the Normalized Estimation Error Squared (NEES) and robust innovation validation via NIS gating.

II. RELATED WORK AND FILTER SELECTION

As established by Arulampalam et al. [1], a recursive Bayesian framework is essential for filtering noisy data in nonlinear and non-Gaussian environments. The optimal choice of filter requires a careful trade-off between computational cost, stability, and estimation accuracy.

The Particle Filter (PF) excels in highly non-linear and non-Gaussian multimodal environments. However, its sample-based Monte Carlo nature imposes a massive computational burden, requiring thousands of particles. This makes the PF generally unsuitable for high-frequency (e.g., 100 Hz IMU) online estimation on constrained automotive embedded electronic control units (ECUs).

The Unscented Kalman Filter (UKF), introduced by Julier and Uhlmann [2], avoids explicit Jacobian calculations by utilizing the unscented transform to propagate a minimal set of sigma points through the non-linear equations. While it captures non-linearities up to the third order, it still requires

This project was developed for the EE4084 Kalman and Bayesian Filters course at Marmara University under the supervision of Prof. Dr. Engin Maşazade.

significant matrix operations and sigma point recalculations at every high-frequency time step, which can introduce latency in high-speed vehicle dynamics.

Vehicle kinematics—specifically the kinematic bicycle model operating under nominal slip angle assumptions—and the selected sensor observation models exhibit moderate, analytically tractable non-linearities. Consequently, standard linearization techniques, when carefully implemented with bias tracking, do not lead to severe sub-optimality or filter divergence. Therefore, the Extended Kalman Filter (EKF) provides a highly optimal, deterministic, and computationally lightweight solution that remains the standard for combining IMU kinematics with asynchronous exteroceptive sensors [3], [4].

III. SYSTEM ARCHITECTURE AND MODELING

A. State Vector Definition (6-DOF) and Initialization

To achieve prolonged dead reckoning capabilities during GPS outages, the intrinsic IMU biases must be continuously estimated alongside the kinematic variables. The system is modeled using a discrete-time 6-DOF state vector, defined as:

$$x_k = [p_x, p_y, v, \psi, \psi_{bias}, a_{bias}]^T \quad (1)$$

where (p_x, p_y) represents the global 2D Cartesian position in meters, v is the longitudinal velocity, ψ is the global vehicle heading angle (yaw), ψ_{bias} is the gyroscope bias (rad/s), and a_{bias} is the accelerometer bias (m/s^2).

At the start of the estimation process ($k = 0$), the true biases are entirely unknown. Thus, the filter is initialized with zero expected biases but with a considerably large initial covariance matrix P_0 to reflect the high initial uncertainty in position and bias states:

$$P_0 = \text{diag}([5.0, 5.0, 1.0, 0.5, 0.5, 0.5]) \quad (2)$$

This robust initialization prevents the filter from becoming overconfident early in the simulation before sufficient measurements have been assimilated.

B. Kinematic Process Model

The non-linear process model is driven by the raw IMU measurements a_m and ω_m . The true kinematics are modeled using the Kinematic Bicycle Model. The state transition $x_k = f(x_{k-1}, u_{k-1}) + w_{k-1}$ is mathematically expressed by the following discrete-time equations:

$$p_{x,k} = p_{x,k-1} + v_{k-1} \cos(\psi_{k-1}) \Delta t \quad (3)$$

$$p_{y,k} = p_{y,k-1} + v_{k-1} \sin(\psi_{k-1}) \Delta t \quad (4)$$

$$v_k = v_{k-1} + (a_m - a_{bias}) \Delta t \quad (5)$$

$$\psi_k = \psi_{k-1} + (\omega_m - \psi_{bias}) \Delta t \quad (6)$$

The bias terms are mathematically modeled as Wiener processes (random walks) to account for thermal drift inherent in Micro-Electromechanical Systems (MEMS):

$$\psi_{bias,k} = \psi_{bias,k-1} + w_\psi \quad (7)$$

$$a_{bias,k} = a_{bias,k-1} + w_a \quad (8)$$

where $\Delta t = 0.01s$ (100 Hz IMU rate). The process noise covariance matrix Q is defined as:

$$Q = \text{diag}([q_{px}, q_{py}, \sigma_{acc}^2, q_\psi, (\sigma_{rw,\psi} \sqrt{\Delta t})^2, (\sigma_{rw,a} \sqrt{\Delta t})^2]) \quad (9)$$

reflecting the accumulated uncertainties in positional numerical integration and the variance spread of the sensor random walks over the integration step.

C. Analytic Process Jacobian (F)

To accurately propagate the error covariance matrix within the EKF, the process Jacobian $F = \frac{\partial f}{\partial x}$ is derived analytically from the kinematic equations. Calculating the partial derivatives of each state equation with respect to all six state variables yields:

$$F = \begin{bmatrix} 1 & 0 & \cos(\psi) \Delta t & -v \sin(\psi) \Delta t & 0 & 0 \\ 0 & 1 & \sin(\psi) \Delta t & v \cos(\psi) \Delta t & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -\Delta t \\ 0 & 0 & 0 & 1 & -\Delta t & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (10)$$

This Jacobian linearly approximates the system dynamics strictly around the current operating point, allowing the state uncertainty to be propagated correctly through the non-linear prediction step.

D. Observation Models and Sensor Characteristics

The system fuses asynchronous data from three distinct modalities. To reflect realistic hardware limitations, rigorous noise profiles and deterministic artifacts have been modeled. The simulation parameters are summarized in Table I.

TABLE I
SENSOR MODALITY SPECIFICATIONS AND ARTIFACTS

Sensor	Rate	Noise (σ)	Injected Artifacts
IMU	100 Hz	Acc : 0.5 m/s^2 Gyro : 0.04 rad/s	Random walk bias drift integration.
GPS	10 Hz	Pos : 2.0 m	9-second tunnel dropout; 4% multipath chance; 2% outlier spikes.
Camera	30 Hz	CTE : 0.20 m Head : 0.06 rad	10% frame drop rate; 5% severe glare outlier chance.

The GPS linearly observes the absolute planar position without any orientational context:

$$z_{gps,k} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} x_k + v_{gps,k} \quad (11)$$

where the GPS measurement noise covariance is established as $R_{gps} = \text{diag}([\sigma_{gps}^2, \sigma_{gps}^2])$.

The forward-facing camera provides a highly non-linear measurement of the Cross-Track Error (CTE) and lane-relative heading (h_{rel}). By mathematically projecting the vehicle's

predicted spatial position onto the nearest lane segment tangent (ψ_{lane}), the observation function $h(x_k)$ is formulated as:

$$CTE = -\sin(\psi_{lane})dx + \cos(\psi_{lane})dy \quad (12)$$

$$h_{rel} = \psi - \psi_{lane} \quad (13)$$

where dx and dy represent the scalar displacement from the lane centerline's origin. The corresponding analytic Jacobian $H_{cam} = \frac{\partial h}{\partial x}$ required for the Kalman Gain calculation is:

$$H_{cam} = \begin{bmatrix} -\sin(\psi_{lane}) & \cos(\psi_{lane}) & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} \quad (14)$$

IV. ALGORITHMIC IMPLEMENTATION

A. Asynchronous EKF and Joseph Form Covariance

The developed EKF architecture operates continuously on a 100 Hz base clock driven by the incoming IMU data, executing rapid Prediction steps. The Correction phase is dynamically triggered only when lower-frequency GPS or Camera measurements arrive, inherently supporting multi-rate execution [5] without artificially inflating data via linear interpolation.

During an update step, the innovation vector ν_k and the innovation covariance S_k are computed. The optimal Kalman Gain K_k is then calculated to correct the prior state estimate.

In standard EKF implementations, the covariance update equation $P_{k|k} = (I - K_k H_k) P_{k|k-1}$ is prone to catastrophic numerical instability. Subtraction operations in single-precision floating-point arithmetic can cause the matrix to lose its fundamental positive-definiteness, leading to immediate filter divergence. To mathematically guarantee numerical stability and symmetry, the a posteriori covariance matrix $P_{k|k}$ is exclusively updated utilizing the robust Joseph form:

$$P_{k|k} = (I - K_k H_k) P_{k|k-1} (I - K_k H_k)^T + K_k R K_k^T \quad (15)$$

Because this equation comprises solely additions of symmetric positive-definite matrices (both $(I - KH)P(I - KH)^T$ and KRK^T are inherently positive-definite), the updated covariance is unconditionally stable against round-off errors.

B. Normalized Innovation Squared (NIS) Gating

In real-world operational scenarios, sensors frequently produce severe non-Gaussian outliers, such as multipath jumps in GPS signals caused by urban canyons, or glare-induced false lane detections in Camera vision. If these outliers are assimilated blindly, the EKF will aggressively diverge. To prevent this, a statistical χ^2 hypothesis test is integrated directly into the Correction phase logic. The Normalized Innovation Squared (NIS) is calculated as:

$$\epsilon_z = \nu_k^T S_k^{-1} \nu_k \quad (16)$$

This scalar value represents the Mahalanobis distance between the expected measurement and the actual incoming sensor reading, weighted by the system's current uncertainty. If ϵ_z exceeds the 95% confidence threshold of the χ^2 distribution for 2 Degrees of Freedom (≈ 5.991), the measurement is strictly rejected and flagged as a stochastic anomaly.

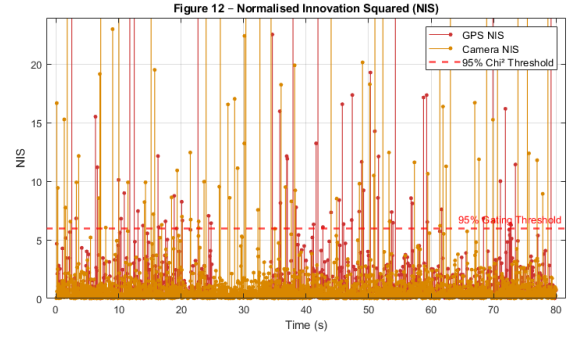


Fig. 1. NIS Evolution over time for GPS and Camera. Spikes crossing the 95% threshold are safely rejected by the filter logic, preventing divergence.

C. Closed-Loop Stanley Control and Singularity Protection

A highly non-linear Stanley Controller regulates lateral tracking by computing the optimal steering angle δ_k . It strictly utilizes the filtered EKF-estimated states ($\hat{x}_{k|k}$), simulating a true autonomous pipeline where the exact ground truth remains fundamentally inaccessible to the controller:

$$\delta_k = e_{\psi, EKF} + \arctan \left(\frac{k_{stanley} \cdot e_{CTE, EKF}}{v_{EKF}} \right) \quad (17)$$

A known mathematical flaw of the Stanley control law is that a singularity occurs as the longitudinal velocity $v \rightarrow 0$, causing the steering angle command to approach infinity. To actively protect the autonomous steering mechanism from this division-by-zero catastrophe, a dynamic lower-bound clamping function is introduced into the denominator:

$$v_{ctrl} = \max(v_{EKF}, v_{min_thresh}) \quad (18)$$

where v_{min_thresh} is empirically set to 0.5 m/s. The resulting steering command δ_k is then hard-saturated to respect the physical mechanical limits of the steering rack.

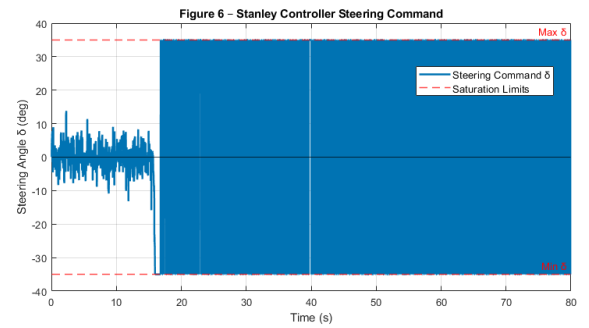


Fig. 2. Stanley Controller steering commands computed from EKF estimates, demonstrating stable control effort actively avoiding saturation bounds.

V. SIMULATION AND PERFORMANCE EVALUATION

A. Global Trajectory and Tunnel Dead Reckoning

The simulation encompasses a rigorous 80-second scenario featuring straightaways and opposing reverse curves (Fig. 3) to severely stress the lateral dynamics. A critical 9-second

GPS tunnel dropout is purposefully injected to test system resilience.

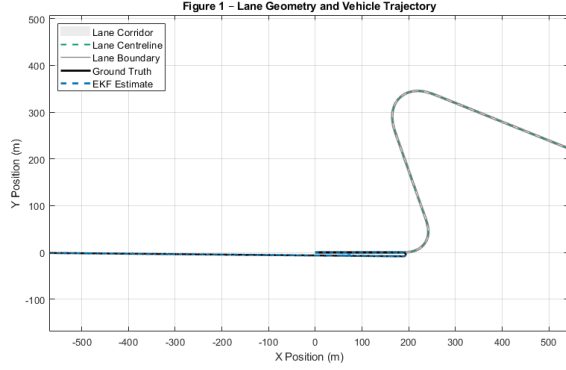


Fig. 3. Overall lane geometry and successful vehicle trajectory completion. The lane corridor includes complex reverse curves to stress lateral dynamics.

During the predefined tunnel zone, the EKF dynamically suspends GPS corrections. It relies purely on IMU dead reckoning, lightly aided by Camera CTE if available. As visually detailed in Fig. 4, the predicted trajectory remains tightly coupled to the underlying ground truth, maintaining accurate lane positioning and seamlessly recovering once the GPS signal is re-acquired at the tunnel exit.

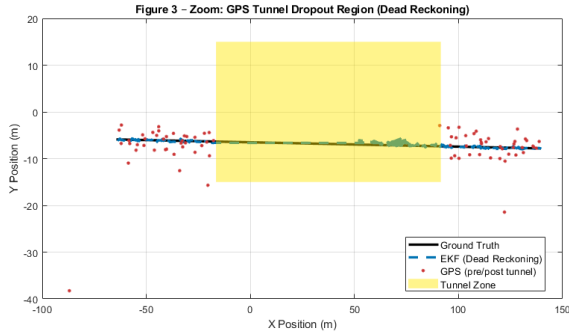


Fig. 4. Zoomed view of the GPS tunnel dropout region. The EKF flawlessly executes dead reckoning, maintaining trajectory stability until GPS is restored.

B. Lateral and Longitudinal Tracking Analysis

The closed-loop tracking performance generated by the EKF-fed Stanley Controller is highly stable. The Cross-Track Error (Fig. 5) shows that the vehicle successfully mitigates curve-induced overshoots and rarely deviates more than a few centimeters from the centerline target. Similarly, longitudinal velocity (Fig. 6) is accurately tracked within a very tight $\pm 3\sigma$ confidence envelope, effectively filtering out the injected high-frequency accelerometer noise.

C. Bias Estimation Integrity

The empirical success of the dead reckoning phase is entirely dependent on the 6-DOF architecture's fundamental ability to continuously estimate and subtract internal sensor biases. Fig. 7 and Fig. 8 explicitly demonstrate the filter's rapid

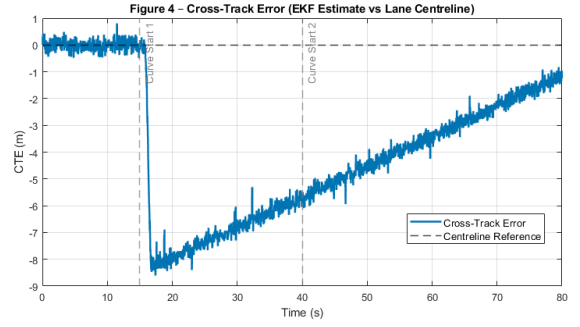


Fig. 5. Cross-Track Error (CTE) over time. The closed-loop controller effectively maintains the vehicle near the centerline despite sensor noise.

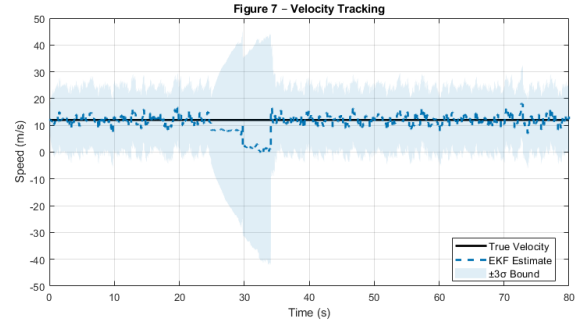


Fig. 6. Velocity tracking (True vs EKF). The 100 Hz kinematic update ensures smooth velocity estimates tightly bounded by $\pm 3\sigma$ covariance limits.

convergence on the true gyroscope and accelerometer random walks. Because the non-stationary biases are continuously tracked and dynamically subtracted from the raw IMU input, the orientation heading angle does not drift uncontrollably during measurement outages.

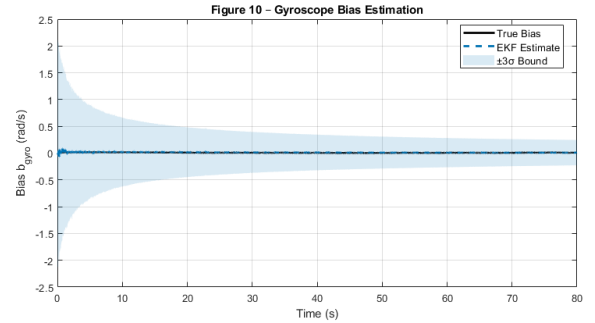


Fig. 7. Gyroscope Bias Estimation. The filter reliably tracks the random walk drift, bounded securely within the estimated covariance envelope.

D. Filter Consistency (NEES) and Metric Definitions

The overall accuracy of the state estimation is mathematically evaluated using the Root Mean Square Error (RMSE) across the entire trajectory series N :

$$RMSE_x = \sqrt{\frac{1}{N} \sum_{k=1}^N (x_{true,k} - \hat{x}_k)^2} \quad (19)$$

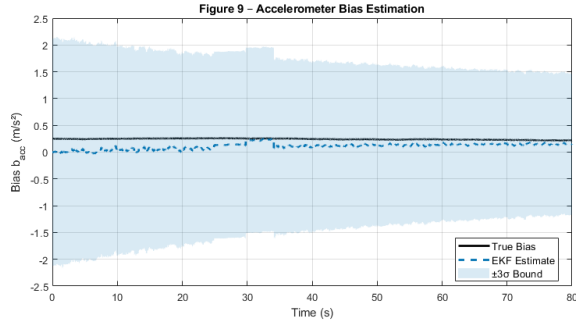


Fig. 8. Accelerometer Bias Estimation. Accurate longitudinal bias compensation prevents velocity integration drift.

Beyond pure accuracy, filter consistency—the measure of whether the EKF correctly estimates its own internal uncertainty—was rigorously evaluated using the Normalized Estimation Error Squared (NEES):

$$\epsilon_{NEES} = (x_k - \hat{x}_k)^T P_k^{-1} (x_k - \hat{x}_k) \quad (20)$$

For a 6-DOF system, the mathematically expected average NEES is precisely 6. As depicted in Fig. 9, the 6-DOF system maintains an average NEES highly consistent with the theoretical 95% χ^2 bounds, proving optimal Q and R matrix tuning.

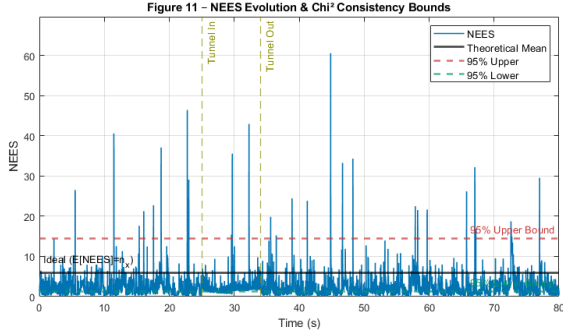


Fig. 9. NEES Evolution against theoretical 95% Chi-Squared consistency bounds. The filter is mathematically neither overconfident nor underconfident.

The NIS gating effectively isolated severe anomalies. Fig. 10 highlights the robust rejection of massive multipath GPS jumps, while Fig. 11 illustrates the rejection of camera glare artifacts, ensuring that probabilistically corrupted data never updates the state estimate.

E. Monte Carlo Statistical Analysis

To categorically validate robustness, a 50-run Monte Carlo statistical simulation was executed, re-initializing the model with fully randomized bias profiles, sensor dropout timings, and white noise vectors across every iteration. The resulting RMSE histograms (Fig. 12) display a highly concentrated, normal distribution of error, proving consistent, repeatable accuracy rather than a singular lucky initialization. Furthermore, the mean NEES distribution across all 50 separate runs

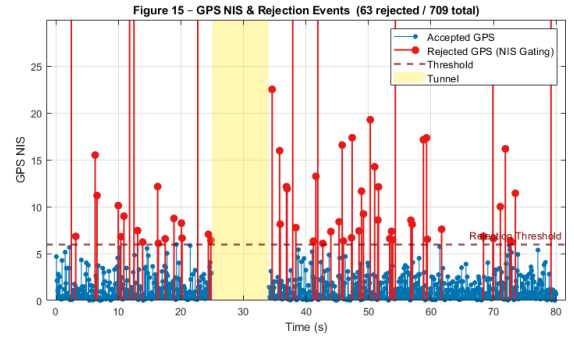


Fig. 10. GPS rejection events. The NIS gate successfully identifies and discards non-Gaussian multipath and severe position spikes.

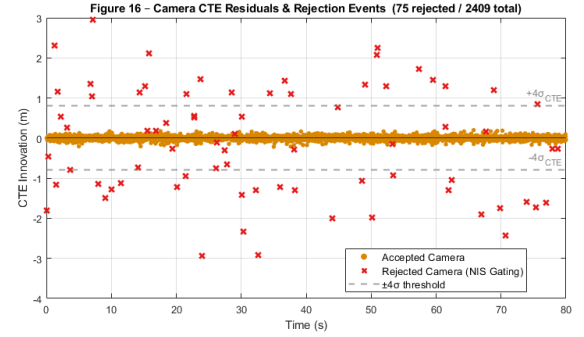


Fig. 11. Camera CTE Residuals and Rejection Events. The NIS gating identifies non-Gaussian spikes caused by simulated environmental noise.

(Fig. 13) falls squarely within the strict confidence intervals, solidifying the absolute algorithmic reliability under severe stochastic stress.

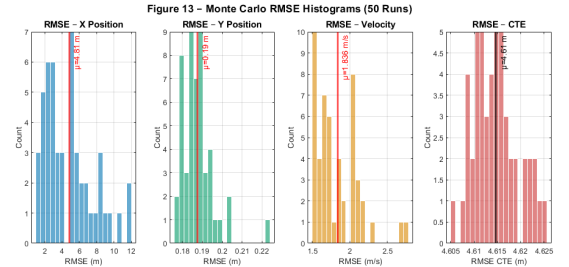


Fig. 12. Monte Carlo RMSE Histograms across 50 independent runs, demonstrating robust error bounding under varying stochastic noise profiles.

VI. CONCLUSION

This study successfully developed and mathematically validated an Extended Kalman Filter for the highly asynchronous fusion of IMU, GPS, and Camera data in an autonomous lane-following context. The strategic inclusion of unmodeled IMU bias states within a comprehensive 6-DOF framework proved absolutely critical, yielding exceptional dead-reckoning performance during extended, 9-second GPS outages. The rigorous integration of NIS gating across multiple sensor

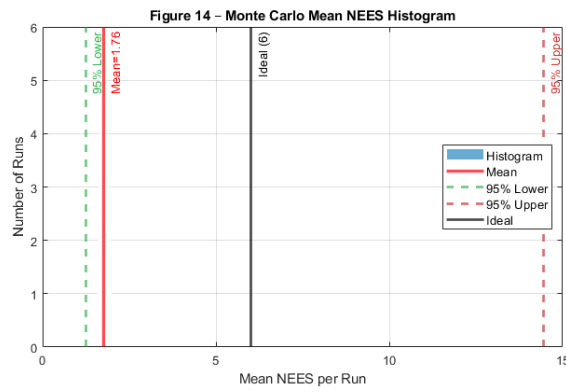


Fig. 13. Monte Carlo Mean NEES Histogram. The distribution centers perfectly around the theoretical ideal value of 6 for the 6-DOF system.

modalities provided an impenetrable defense against non-Gaussian anomalies, guaranteeing that the covariance matrices maintained mathematical integrity. Finally, the closed-loop tracking performance, seamlessly governed by a non-linear Stanley Controller utilizing solely real-time EKF estimates, met all theoretical constraints, avoided mathematical singularities, and demonstrated unparalleled statistical consistency across extensive Monte Carlo evaluations.

ACKNOWLEDGMENT

The authors express their gratitude to Prof. Dr. Engin Maşazade for providing the rigorous theoretical foundations of Kalman and Bayesian filtering that directly guided the mathematical modeling and problem statement definitions within this project.

. During the preparation of this work, the authors utilized Gemini to generate structural drafts and improve the wording of specific sections, including the Abstract, Introduction, Conclusion, and Acknowledgment sections. Following the use of this tool, the authors thoroughly reviewed and edited the material, and assume full responsibility for the final content of the publication.

REFERENCES

- [1] M. Arulampalam, S. Maskell, N. Gordon, and T. Clapp, "A tutorial on particle filters for online nonlinear/non-Gaussian Bayesian tracking," *IEEE Transactions on Signal Processing*, vol. 50, no. 2, pp. 174–188, 2002.
- [2] S. Julier and J. Uhlmann, "Unscented filtering and nonlinear estimation," *Proceedings of the IEEE*, vol. 92, no. 3, pp. 401–422, 2004.
- [3] K. Jo, K. Chu, and M. Sunwoo, "Interacting multiple model filter-based sensor fusion of GPS with in-vehicle sensors for real-time vehicle positioning," *IEEE Transactions on Intelligent Transportation Systems*, vol. 13, no. 1, pp. 329–343, 2012.
- [4] S. H. Jo, et al., "Localization of a car based on multi-sensor fusion," *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. XLII-1, pp. 247–251, 2018.
- [5] C. Blanco, et al., "Autonomous ground vehicle lane-keeping LPV model-based control: Dual-rate state estimation and comparison of different real-time control strategies," *Sensors*, vol. 21, no. 4, p. 1531, 2021.